

Marks :- 10

Fall-Semester 2025-26
Digital Assignment - II

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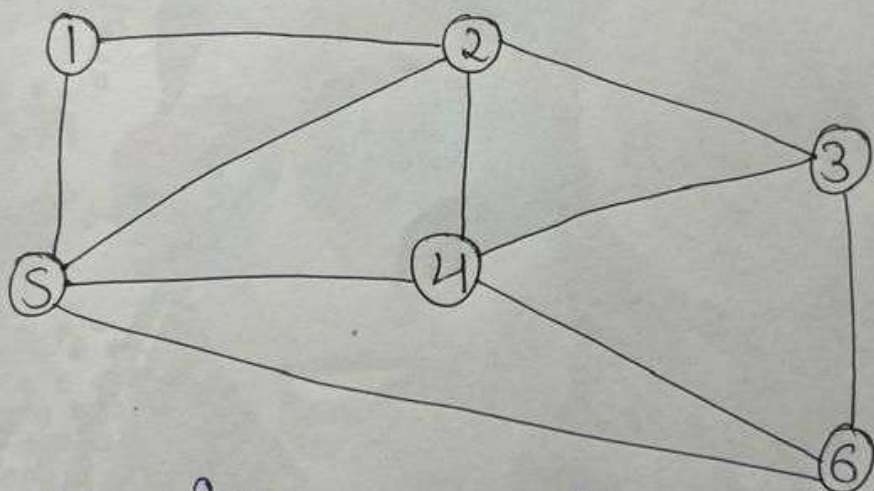
Course code :- BCSE202L

Program/Branch :- BTech

Course Name :- Data Structure And Algorithm

Faculty Name :- Dr. Prabim S.M

Ques-1 Determine the DFS and BFS traversal for the following graph :



Solve - ① Starting node - we will start the traversal at node 1.

First, let's list the neighbours for each node (the Adjacency list) in ascending order, based on the graph.

1: {2, 5}

2: {1, 3, 4, 5}

3: {2, 4, 6}

4: {2, 3, 5, 6}

5: {1, 2, 4, 6}

6: {3, 4, 5}

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① Depth first Search :- (DFS) - It explores as far as possible down one branch before backtracking. It uses a stack.

→ Step by step traversal :-

1. Start at 1: visit 1

- Traversal: [1]

- visited: [1]

2. Explore Neighbours of 1 (2, 5): Go to the first unvisited neighbour, 2, visit (2).

- Traversal: [1, 2]

- visited: [1, 2]

3. Explore Neighbour of 2 (1, 3, 4, 5):

- 1 is already visited

- Go to the next unvisited neighbour 3, visit

- Traversal: [1, 2, 3]

- visited: [1, 2, 3]

4. Explore Neighbours of 3 (2, 4, 6) : (3)
- 2 already visited
 - Go to the next unvisited neighbour 4, visit 4.
 - Traversal : [1, 2, 3, 4]
 - Visited : [1, 2, 3, 4]

5. Explore Neighbours of 4 (2, 3, 5, 6) :
- 2 and 3 already visited
 - Go to the next traversal neighbour 5, visit 5.
 - Traversal : [1, 2, 3, 4, 5]
 - Visited : [1, 2, 3, 4, 5]

6. Explore Neighbours of 5 (1, 2, 4, 6) :
- 1, 2 and 4 already visited
 - Go to the unvisited neighbour 6, visit 6.
 - Traversal : [1, 2, 3, 4, 5, 6]
 - Visited : [1, 2, 3, 4, 5, 6]

7. Explore Neighbours of 6 (3, 4, 5) :
- All neighbours are Already visited
 - Backtrack to node 5.

8. Backtrack to 5 (1, 2, 4, 6) :
- All neighbours are Already visited
 - Back track to node 4.

9: Backtrack to 4 (2, 3, 5, 6):

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- All neighbour already visited
- Backtrack to node 3

10: Backtrack to 3 (2, 4, 6):

- All neighbours are already visited
- Backtrack to node 2.

11: Backtrack to 2 (1, 3, 4, 5):

- All neighbours are already visited
- Backtrack to node 1.

12: Backtrack to 1 (2, 5):

- All neighbours are Already visited
- Backtrack to node But all node visited.

Final DFS : 1, 2, 3, 4, 5, 6 }
traversal

② Breadth first Search (BFS) :-

- BFS explores all neighbour at the current "level" before moving on to the next level. It uses a Queue (first-in, first-out).

Step by Step traversal :-

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Step	Action	Queue	Visited Set	Final traversal
1.	Start : Add 1 to Queue. Mark 1 as visited	[1]	{1}	[]
2.	Dequeue 1. Add to traversal. Get neighbour of 1 {2, 5}	[2, 5]	{1, 2, 5}	[1]
	Add 2 and 5 to Queue. Mark them visited	[5, 3, 4]	{1, 2, 5, 3, 4}	[1, 2]
3.	Dequeue 2 : Add to traversal. Get neighbour {1, 3, 4, 5}	[3, 4, 6]	{1, 2, 5, 3, 4, 6}	[1, 2, 5]
4.	Dequeue 5 :	[4, 6]	{1, 2, 5, 3, 4, 6}	[1, 2, 5, 3]
5.	Dequeue 3 :	[6]	{1, 2, 5, 3, 4, 6}	{1, 2, 5, 3, 4}
6.	Dequeue 4 :	[]	{1, 2, 5, 3, 4, 6}	[1, 2, 5, 3, 4, 6]
7.	Dequeue 6 :			
8.	Queue is empty : Traversal is completed		BFS → traversal	[1, 2, 5, 3, 4, 6]

Ques-2 (A) Given the Array $[7, 2, 1, 6, 8, 5, 3, 4]$, what is the final state of the array after all partition is complete (using the homomoto partition scheme)? Show all the Step-by-step process with a clean Diagram. (6)

(B) Consider the following 10-element array:

$$A = [8, 3, 10, 1, 6, 4, 2, 9, 7, 5]$$

Trace the Insertion Sort algorithm on array. Show the state of the array after each pass (after each element from index 1 to 9 is inserted into its correct position in the sorted subarray).

Solve - (A) $[7, 2, 1, 6, 8, 5, 3, 4]$

$$x = 7, i = 0, j = 1$$

$$\textcircled{1} \quad \begin{array}{ccccccc} [7, & 2, & 1, & 6, & 8, & 5, & 3, & 4] \\ i & j \end{array}$$

$$A[j] \leq x \quad (2 < 7)$$

$$i = 1$$

$$\text{Swap } [A[i], A[j]]$$

$$j = 2$$

$$\textcircled{2} [7, 2, 1, 6, 8, 5, 3, 4]$$

⑦

$$A[i] \leq x \quad (1 \leq 7)$$

$$i = 2$$

$$\text{swap}(A[2], A[2])$$

$$j = 3$$

$$\textcircled{3} [7, 2, 1, 6, 8, 5, 3, 4]$$

$$A[i] \leq x \quad (6 \leq 7)$$

$$i = 3$$

$$\text{swap}(A[3], A[3])$$

$$j = 4$$

$$\textcircled{4} [7, 2, 1, 6, 8, 5, 3, 4]$$

$$A[i] \leq 7$$

$$j = 5$$

$$\textcircled{5} [7, 2, 1, 6, 8, 5, 3, 4]$$

$$A[i] \leq 7 \quad (5 \leq 7)$$

$$i = 4$$

$$\text{swap}(A[4], A[5])$$

$$j = 6$$

$$\textcircled{6} [7, 2, 1, 6, 5, 8, 3, 4]$$

$$A[j] \leq 7 \quad (5 \leq 7)$$

$$i = 5$$

$$\text{Swap}(A[5], A[6])$$

$$j = 7$$

Now the subarray part to be continue:-

$$\rightarrow [4, 2, 1, 6, 5, 3, \boxed{7}, 8]$$

$$\text{Subarray } [4, 2, 1, 6, 5, 3]$$

$$x = 4, i = 0, j = 1$$

$$\textcircled{1} \begin{matrix} i & j \\ [4, 2, 1, 6, 5, 3] \end{matrix}$$

$$A[j] \leq 2 \quad (2 \leq 4)$$

$$i = 1$$

$$\text{Swap}(A[i], A[j])$$

$$j = 2$$

$$\textcircled{2} \begin{matrix} i & j \\ [4, 2, 1, 6, 5, 3] \end{matrix}$$

$$A[j] \leq 4 \quad (1 \leq 4)$$

$$i = 2$$

$$\text{Swap}(A[i], A[j])$$

$$j = 3$$

$$\textcircled{3} [4, 2, \underset{i}{1}, \underset{j}{6}, 5, 3]$$

$$A[j] \leq 4 \quad (6 \leq 4)$$

$$j = 4$$

$$\textcircled{4} [4, 2, \overset{(i)}{1}, 6, \overset{(j)}{5}, 3]$$

$$A[j] \leq 4 \quad (5 \leq 4)$$

$$j = 5$$

$$\textcircled{5} [4, 2, \underset{i}{1}, \underset{(i)}{6}, \underset{(j)}{5}, 3]$$

$$A[j] \leq 4 \quad (3 \leq 4)$$

$$\text{swap}(6, 3)$$

$$j = 6$$

$$\textcircled{6} [4, 2, 1, 3, 5, 6]$$

$$\textcircled{7} [3, 2, 1, \boxed{4}, 5, 6]$$

$$\text{Subarray } [3, 2, 1]$$

$$\left. \begin{array}{l} k = 3 \\ p = 0 \\ q = 1 \end{array} \right\}$$

$$① \begin{bmatrix} 3, 2, 1 \\ i, j \end{bmatrix}$$

$$A[j] \leq 3 \quad (2 \leq 3)$$

$$i = 1$$

$$\text{swap}(A[i], A[j])$$

$$j = 2$$

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$$② \begin{bmatrix} 1, 2, 3 \\ i, j \end{bmatrix}$$

$$A[j] \leq 3 \quad (1 \leq 3)$$

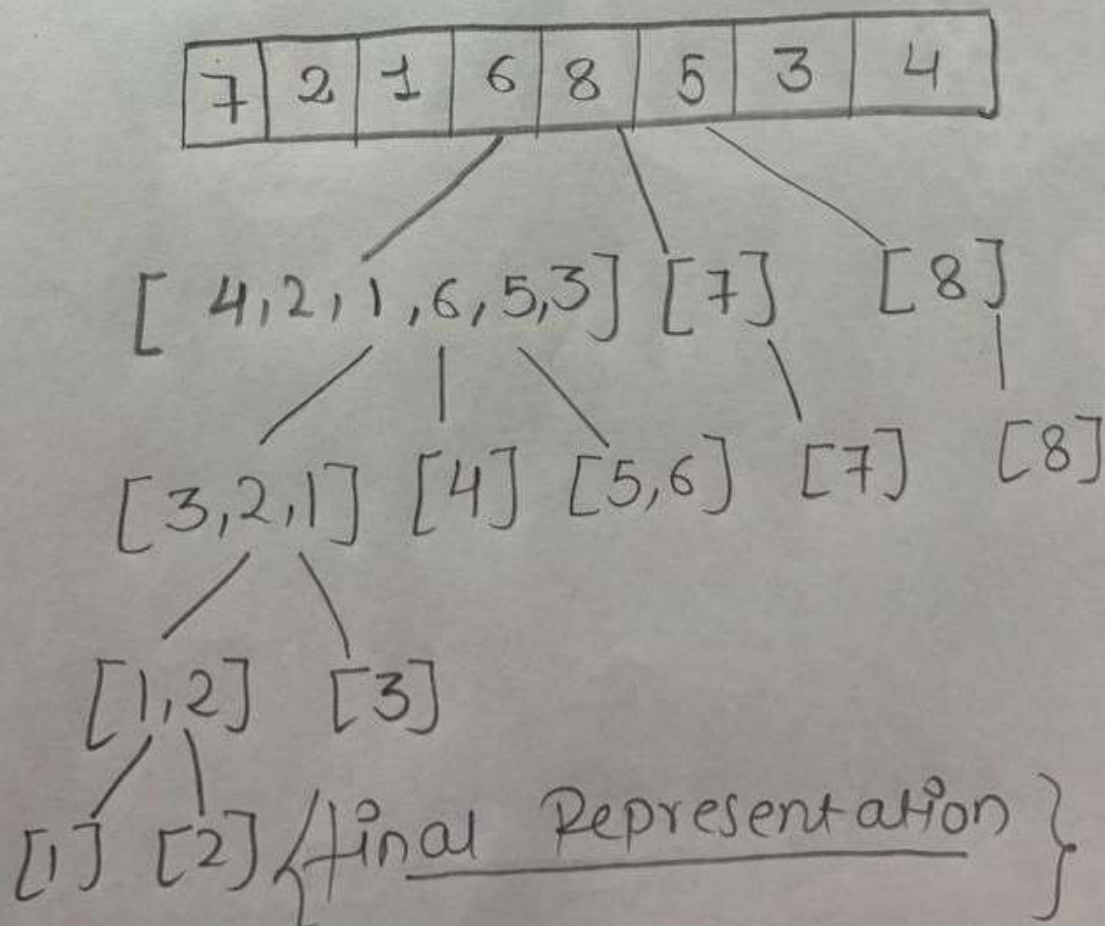
$$i = 2$$

$$\text{swap}(A[i], A[j])$$

$$j = 3$$

{now Already sorted Array}

2) Graph Representation of Question :-



Part - (B) $A = [8, 3, 10, 1, 6, 4, 2, 9, 7, 5]$ (11)

$$i = 9, j = 1, \text{key} = 3$$

$$i > 0 \text{ and } A[i] > 3 \quad (0 > 0 \text{ and } 8 > 3)$$

$$A[i] = A[0]$$

$$i = -1$$

$$A[0] = 3$$

$$A = [3, 8, 10, 1, 6, 4, 2, 9, 7, 5]$$

$$i = 1, j = 2, \text{key} = 10$$

$$i \geq 0 \text{ and } A[i] > 10 \quad (1 \geq 0 \text{ and } 8 > 10) \text{ false}$$

$$A[2] = 10$$

$$A = [\overset{1}{\cancel{3}}, \overset{3}{\cancel{8}}, \overset{8}{\cancel{10}}, \overset{10}{\cancel{1}}, 6, 4, 2, 9, 7, 5]$$

$$i = 2, j = 3, \text{key} = 1$$

$$i \geq 0 \text{ and } A[i] > \text{key} \quad (2 \geq 0 \text{ and } 10 > 1)$$

$$A[3] = A[2]$$

$$i = 1$$

$$i \geq 0 \text{ and } A[i] > \text{key} \quad (1 \geq 0 \text{ and } 8 > 1)$$

$$i = 0$$

$$0 \geq 0 \text{ and } 3 > 1$$

$$A[1] = A[0]$$

$$j = -1$$

$$A = [1, 3, \overset{6}{\cancel{8}}, \overset{8}{\cancel{10}}, \overset{10}{\cancel{6}}, 4, 2, 9, 7, 5]$$

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$$i = 3, j = 4, \text{key} = 6$$

$$i \geq 0 \text{ and } A[i] > \text{key} \quad (3 \geq 0 \text{ and } 10 > 6)$$

$$A[4] = A[3]$$

$$i = 2$$

$$2 \geq 0 \text{ and } 8 > 6$$

$$(A[3] = A[2])$$

$$i = 1$$

$$(1 \geq 0 \text{ and } 3 > 6) \text{ false}$$

$$A[2] = \text{key}$$

$$A = [1, 3, 6, \overset{6}{\cancel{8}}, \overset{8}{\cancel{10}}, \overset{10}{\cancel{4}}, 2, 9, 7, 5]$$

$$i = 4, j = 5, \text{key} = 4$$

$$i \geq 0 \text{ and } 10 > 4 \quad (4 \geq 0 \text{ and } 10 > 4)$$

$$A[5] = A[4]$$

$$i = 3$$

$$3 \geq 0 \text{ and } 8 > 4$$

$$A[4] = A[3]$$

$$i = 2$$

$$i = 2$$

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(2 > 0 and 8 > 4)

$$A[3] = A[2]$$

(1 > 0 and 3 > 4)

$$A[2] = 4$$

$$A = [1, 2, 3, 4, 5, 6, 8, 9, 10, 7, 5]$$

$$i = 6, j = 7, \text{key} = 9$$

(6 > 0 and 10 > 9)

$$A[7] = A[6]$$

$$i = 5$$

(5 > 0 and 8 > 9)
false

$$A[6] = \text{key}$$

$$A = [1, 2, 3, 4, 6, 7, 8, 9, 10, 5]$$

$$i = 7, j = 5, \text{key} = 7$$

7 > 0 and 10 > 7

$$A[8] = A[9]$$

$$i = 6$$

$6 > 0$ and $9 > 4$

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$$A[7] = A[6]$$

$$i = 5$$

$5 > 0$ and $8 > 7$

$$A[6] = A[5]$$

$$\text{arr}[6] = \text{key}$$

$$A = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$$

$$i = 8, j = 9, \text{key} = 5$$

$8 > 0$ and $10 > 5$

$$A[9] = A[8]$$

$$i = 7$$

$7 > 0$ and $9 > 9$

$$A[8] = A[7]$$

$6 > 0$ and $8 > 5$

$$A[7] = A[6]$$

$A[5] > 0$ and $7 > 5$

$$A[6] = A[5]$$

$4 > 0$ and $6 > 5$

$$A[5] = A[4]$$

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$A[4] = \text{Key}$

Final Sorted Array $\Rightarrow A = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$

Ques-3 Insert the keys $X [K, R, P, C, S, N, Y, T, J, M]$

into hash table of size 10 using the

hash function $h(x) = (\text{ord}(x) - \text{ord}('A') + 1) \% 10$

Assume $\text{ord}('A') \Rightarrow 65$

(i) Display the resulting hash table and also specify the number of collisions.

(ii) Apply linear probing, quadratic probing and show the resultant hash table.